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Inequalities

Linear: flip the sign when you divide by a negative. Quadratic: critical values, sketch, choose. Regions: one inequality per edge.

THE TOPIC ON ONE MAP

Inequalities

01 — Linear inequalities

÷ negative: flip set notation

02 — Quadratic inequalities

roots → sketch → read “satisfied by both”

03 — Inequalities for k

no real roots real roots

04 — Regions on a graph

define R above / below

01 Linear Inequalities

Exactly like an equation — with one rule: multiplying or dividing by a **negative** flips the sign.

WHEN YOU SEE → YOU DO

“solve $3x + 4 \leq 5(x - 1)$ ”

→ expand, collect, divide

dividing by a negative

→ flip: $>$ becomes $<$

“in set notation”

→ $\{x : x \geq 3\} \cdot \{x : -1 < x \leq 4\}$

$\frac{4}{z} < 8$ — unknown on the bottom

→ multiply by z^2 (always positive), never by z

WORKED MODEL • JANUARY 2026

$$\frac{4}{z} < 8$$

1 $4z < 8z^2 \Rightarrow 8z^2 - 4z > 0$

× z^2 , sign safe

2 $4z(2z - 1) > 0$, critical values $0, \frac{1}{2}$

factorise

ANSWER $z < 0$ or $z > \frac{1}{2}$

IDEA 01

Practise: Linear Inequalities

My questions first, then selected practice.

Q1 **DR ESLAM** Flip the sign

P1 · TOPIC 4 2 marks

Solve (a) $3x - 2 > 7$ (b) $5 - 2x \leq 11$

SELECTED PRACTICE**Q2** **PRACTICE** Practice · Easy

P1 · TOPIC 4 3 marks

1 Solve the inequalities:

(i) $2x \geq 8$

(ii) $3 + 2x < 11$

(iii) $5 + x > 4x - 1$

(3 marks)

Q3**PRACTICE**

Practice · Easy

P1 · TOPIC 4**4 marks**

2 Solve the inequalities:

(i) $2x - 9 \geq 5(x - 3)$

(ii) $3(5 - x) < 2(9 - 2x)$

(4 marks)**Q4****PRACTICE**

Practice · Medium

P1 · TOPIC 4**3 marks**

1 Solve the inequality $3x + 4 \leq 5(x - 1)$.

(3 marks)

02 Quadratic Inequalities

Three steps: everything on one side, critical values (calculator), sketch the cup and read off.

WHEN YOU SEE → YOU DO

“> 0” with a cup shape

→ outside the roots: $x < \alpha$ or $x > \beta$

“< 0” with a cup shape

→ between the roots: $\alpha < x < \beta$

“using algebra”, “not calculator technology”

→ show the factorising or formula

two inequalities “satisfied by both”

→ number line, take the overlap

1

One side = 0

Rearrange so the x^2 term is positive.

2

Critical values

Solve the equation (calculator first).

3

Sketch and choose

Above the axis (> 0) = outside · below (< 0) = inside.

WORKED MODEL · JUNE 2023

$$4x^2 - 3x + 7 \geq 4x + 9$$

1 $4x^2 - 7x - 2 \geq 0$

one side

2 $(4x + 1)(x - 2) \geq 0$, $x = -\frac{1}{4}, 2$

critical values

ANSWER $x \leq -\frac{1}{4}$ or $x \geq 2$

IDEA 02

Practise: Quadratic Inequalities

My questions first, then selected practice, then real past papers.

Q5 **DR ESLAM** Inside the roots

P1 · TOPIC 4 3 marks

Solve $x^2 - x - 12 \leq 0$.

Q6 **DR ESLAM** Outside the roots

P1 · TOPIC 4 3 marks

Solve $x^2 - 4x + 3 > 0$.

SELECTED PRACTICE

Q7

PRACTICE

Practice · Easy

P1 · TOPIC 4 · 6 marks

3 (a) Write down the solutions to $(x - 3)(x - 8) = 0$.

(2 marks)

(b) Sketch the graph of $y = (x - 3)(x - 8)$, clearly showing the coordinates of the points where the graph intercepts the x -axis.

(2 marks)

(c) Hence, or otherwise, solve the inequality $(x - 3)(x - 8) < 0$.

(2 marks)

Q8**PRACTICE**

Practice · Medium

P1 · TOPIC 4**5 marks**

5 Find the values of x that satisfy the inequalities

$$x^2 + 3x > 4$$

$$4x + 1 > 4$$

(5 marks)**Q9****PRACTICE**

Practice · Very Hard

P1 · TOPIC 4**4 marks**

1 Solve the simultaneous inequalities

$$t^2 - 2t - 15 < 0 \text{ and}$$

$$t^2 + 14 \leq 9t.$$

(4 marks)

PAST PAPERS

Q10 **PAST PAPER** Pearson IAL WMA11 • June 2023 • Q1**P1 • TOPIC 4** **4 marks**

1. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Solve the inequality

$$4x^2 - 3x + 7 \geq 4x + 9$$

(4)

(Total for Question 1 is 4 marks)

03 Inequalities for k : the Discriminant

A condition on the number of roots becomes a quadratic inequality in k .

WHEN YOU SEE → YOU DO

“does not cross or touch”, “no real roots”

$$\longrightarrow b^2 - 4ac < 0$$

“two distinct real roots”

$$\longrightarrow b^2 - 4ac > 0$$

“real roots”

$$\longrightarrow b^2 - 4ac \geq 0$$

a line and a curve

$$\longrightarrow \text{line} = \text{curve} \rightarrow \text{one quadratic} \rightarrow \text{discriminant}$$

WORKED MODEL • JANUARY 2020

$y = k(2x - 1)$ does not cross or touch $y = x^2 + 2x + 11$.

1 $x^2 + (2 - 2k)x + 11 + k = 0$

curve = line

2 $(2 - 2k)^2 - 4(11 + k) < 0 \Rightarrow k^2 - 3k - 10 < 0$

discriminant < 0

ANSWER $-2 < k < 5$

IDEA 03

Practise: Inequalities for k

My questions first, then selected practice, then real past papers.

Q11 DR ESLAM No real roots

P1 · TOPIC 4 3 marks

$x^2 + 2x + k = 0$ has no real roots. Find the range of k .

SELECTED PRACTICE**Q12** PRACTICE Practice · Easy

P1 · TOPIC 4 3 marks

7 (a) Write down, in terms of k , the discriminant of $x^2 + 8x + 4k$.

(1 mark)

(b) Hence find the values of for k which the equation $x^2 + 8x + 4k = 0$ has two real and distinct solutions.

(2 marks)

Q13**PRACTICE**

Practice · Medium

P1 · TOPIC 4 4 marks

- 3 The equation $kx^2 + 2kx + 4 = 0$, where k is a constant, has two distinct real roots.

Find the possible value(s) of k .

(4 marks)**Q14****PRACTICE**

Practice · Hard

P1 · TOPIC 4 4 marks

- 3 The equation $(kx)^2 + (k - 2)x + 1 = 0$, where k is a constant, has two distinct real roots. Find the possible values of k .

(4 marks)

PAST PAPERS

Q15 PAST PAPER Pearson IAL WMA11 · January 2020 · Q8

P1 · TOPIC 4 6 marks

8. The straight line l has equation $y = k(2x - 1)$, where k is a constant.

The curve C has equation $y = x^2 + 2x + 11$

Find the set of values of k for which l does not cross or touch C .

(6)

(Total 6 marks)

04 Regions on a Graph

Every edge of the region is one inequality. Solid line $\leq/\geq$, dashed line $</>$.

WHEN YOU SEE → YOU DO

“use inequalities to define R ”

→ one inequality per boundary

region below a line/curve

→ $y \leq \dots$

region above

→ $y \geq \dots$

a vertical edge $x = a$

→ $x \geq a$ or $x \leq a$

“represent $x + 5 \leq y \leq x(x - 3)$ graphically”

→ shade where the curve is above the line

WORKED MODEL • JUNE 2023

R_1 : $0 \leq y \leq 2$, $y \leq 10 - 2x$, $y \leq kx$; area $\frac{27}{4}$; $x = a$ through $y = 2$ on $y = kx$.

1 trapezium: $\frac{1}{2} \cdot 2 \cdot (5 + (4 - a)) = \frac{27}{4}$

area

2 $a = \frac{9}{4}$, $k = \frac{2}{a} = \frac{8}{9}$

solve

ANSWER R_2 : $x \geq \frac{9}{4}$, $y \geq \frac{8}{9}x$, $y \leq 10 - 2x$

IDEA 04

Practise: Regions on a Graph

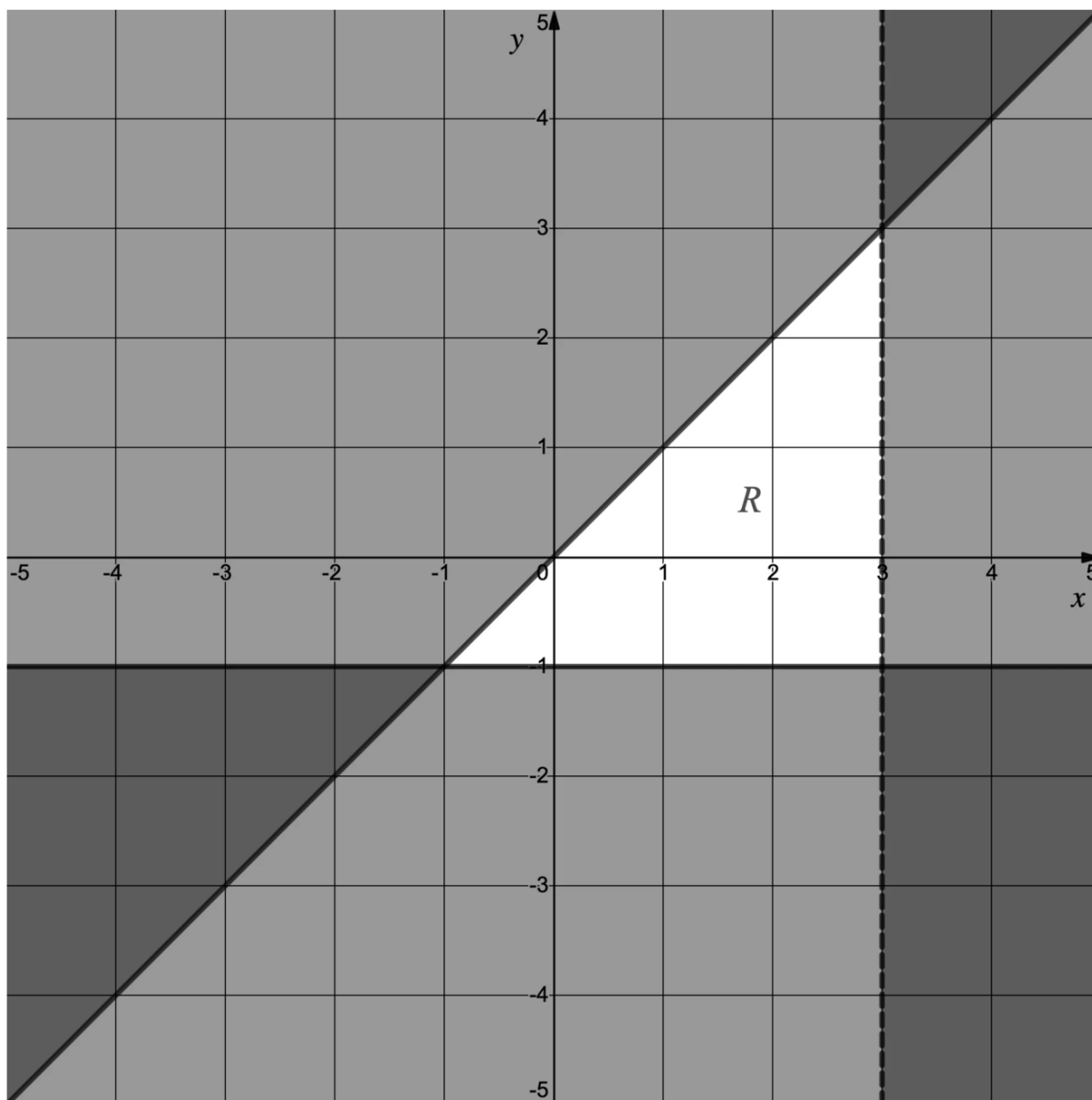
My questions first, then selected practice, then real past papers.

SELECTED PRACTICE

Q16 PRACTICE Practice · Easy

P1 · TOPIC 4 3 marks

- 8 Write down the three inequalities that define the region R shown in the diagram below.



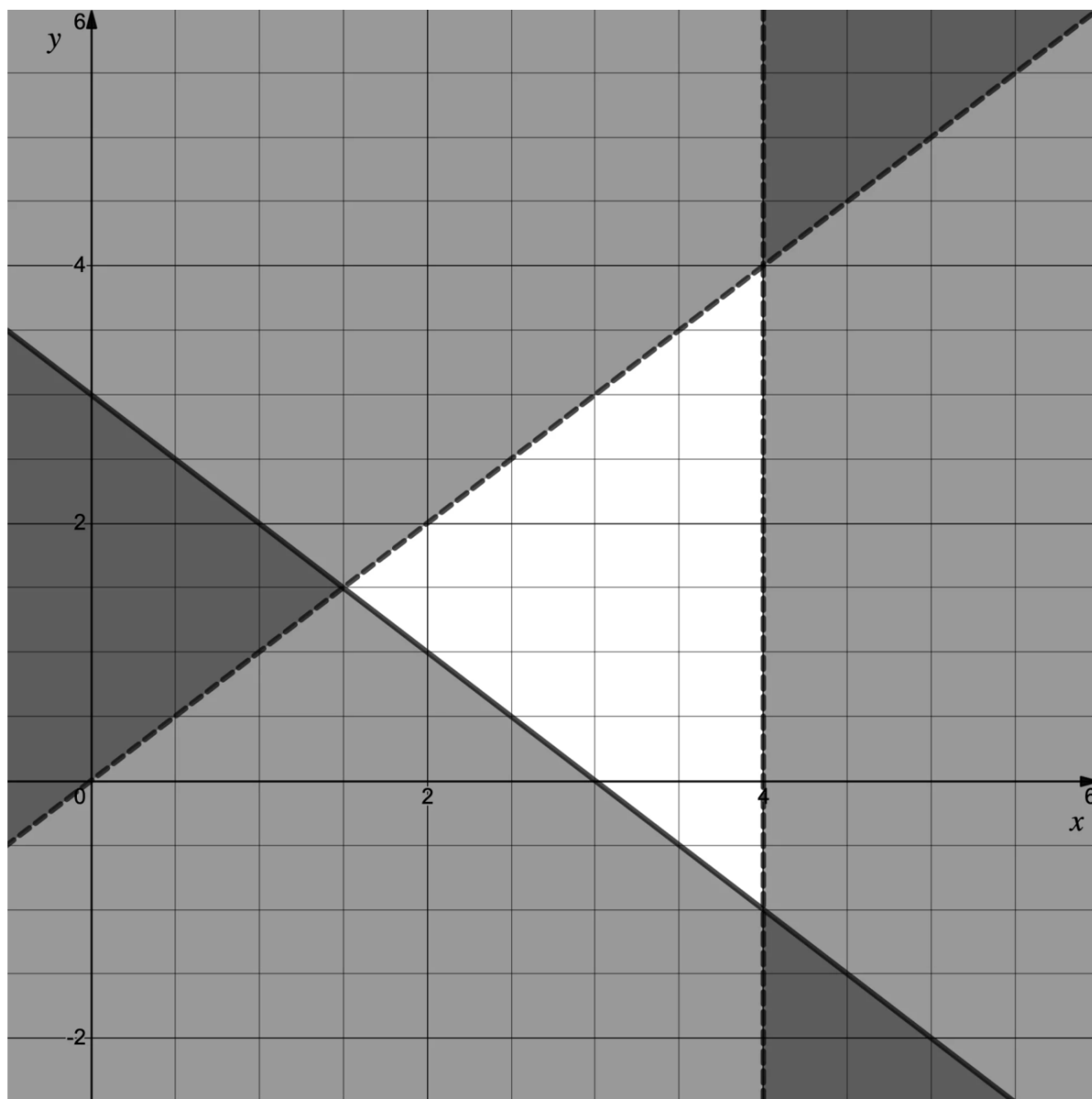
(3 marks)

Q17**PRACTICE**

Practice • Medium

P1 • TOPIC 4 4 marks

8 Write down the inequalities that define the region R shown in the diagram below.



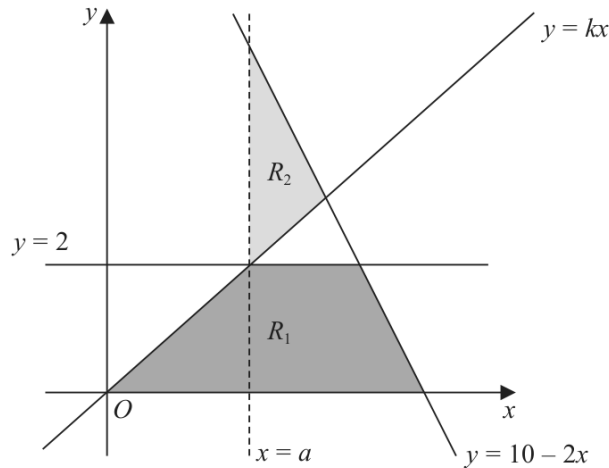
(4 marks)

PAST PAPERS

Q18 PAST PAPER Pearson IAL WMA11 · June 2023 · Q7

P1 · TOPIC 4 6 marks

7.

**Figure 2**

The region R_1 , shown shaded in Figure 2, is defined by the inequalities

$$0 \leq y \leq 2 \quad y \leq 10 - 2x \quad y \leq kx$$

where k is a constant.

The line $x = a$, where a is a constant, passes through the intersection of the lines $y = 2$ and $y = kx$

Given that the area of R_1 is $\frac{27}{4}$ square units,

(a) find

- (i) the value of a
- (ii) the value of k

(4)

(b) Define the region R_2 , also shown shaded in Figure 2, using inequalities.

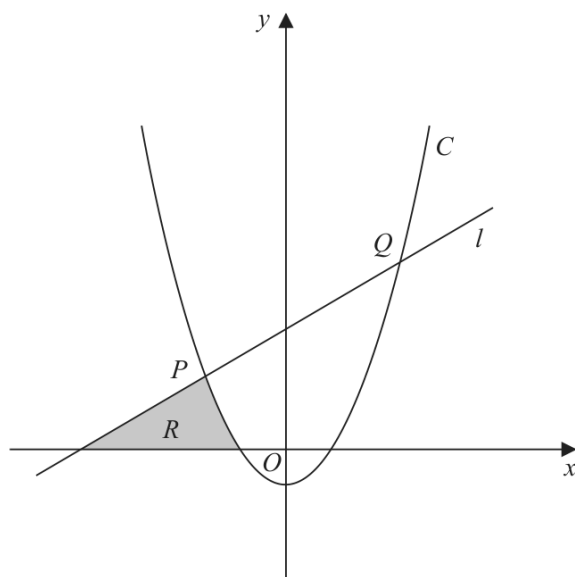
(2)**(Total for Question 7 is 6 marks)**

Q19**PAST PAPER**

Pearson IAL WMA11 • June 2024 • Q6

P1 • TOPIC 4 7 marks

6.

**Figure 3**

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Figure 3 shows

- the line l with equation $y - 5x = 75$
- the curve C with equation $y = 2x^2 + x - 21$

The line l intersects the curve C at the points P and Q , as shown in Figure 3.

- (a) Find, using algebra, the coordinates of P and the coordinates of Q .

(4)

The region R , shown shaded in Figure 3, is bounded by C , l and the x -axis.

- (b) Use inequalities to define the region R .

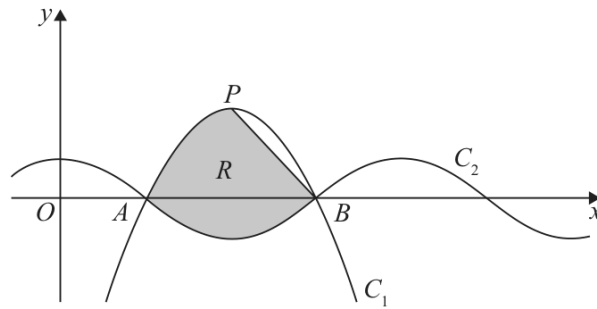
(3)

(Total for Question 6 is 7 marks)

Q20 PAST PAPER Pearson IAL WMA11 · January 2025 · Q9

P1 · TOPIC 4 13 marks

9.

**Figure 2**

- (a) Express $6x - \frac{27}{4} - x^2$ in the form $a + b(x + c)^2$ where a , b and c are constants to be found.

(3)

Figure 2 shows part of a sketch of curve C_1 with equation

$$y = 6x - \frac{27}{4} - x^2$$

Given that the point P is the maximum point on C_1

- (b) state the coordinates of P

(2)

Figure 2 also shows part of a sketch of curve C_2 with equation

$$y = \cos(kx)$$

where k is a constant and x is measured in radians.

Given that C_1 and C_2 intersect on the x -axis at point A and at point B , as shown in Figure 2,

- (c) (i) state the x coordinate of B
 (ii) state the value of k
 (iii) state the period of C_2

(3)

The line segment L joins P and B .

The region R , shown shaded in Figure 2, is bounded by L , C_1 and C_2

- (d) Use inequalities to define R .

(5)

(Total for Question 9 is 13 marks)

Q21 PAST PAPER Pearson IAL WMA11 • January 2026 • Q3

P1 • TOPIC 4 5 marks

3. (i) Find the range of values of
- z
- for which

$$\frac{4}{z} < 8$$

giving the answer in simplest form.

(2)

- (ii)

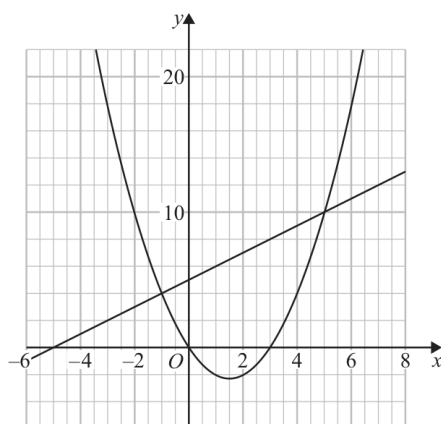


Figure 1

Figure 1 shows a plot of

- the line with equation $y = x + 5$
- the curve with equation $y = x(x - 3)$

On the opposite page there is a copy of Figure 1 labelled Diagram 1.

On Diagram 1 represent the following inequality graphically.

$$x + 5 \leq y \leq x(x - 3) \quad \text{for } x \geq -2$$

(3)

Question 3 continued

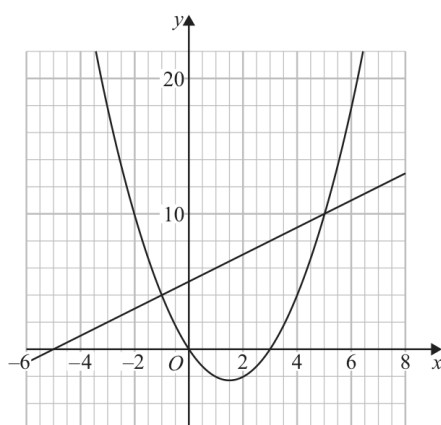


Diagram 1

(Total for Question 3 is 5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|---------------------------|---|
| 01 | Linear | Like an equation; negative divide flips. Unknown on the bottom: $\times$ square. |
| 02 | Quadratic | One side $\rightarrow$ critical values $\rightarrow$ sketch: > 0 outside, < 0 inside. |
| 03 | For k | Number of roots $\rightarrow$ discriminant $\rightarrow$ quadratic inequality in k . |
| 04 | Regions | One inequality per edge. Solid $\leq$, dashed $<$. |

BOTTOM LINE Critical values $\rightarrow$ sketch $\rightarrow$ read the region. Never divide by an unknown whose sign you don't know.

Answers

Inequalities • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • LINEAR INEQUALITIES

- | | | |
|-----------|--------------------------|---|
| Q1 | Dr Eslam • Flip the sign | (a) $x > 3$ (b) $x \geq -3$ |
| Q2 | Practice • Easy | (i) $x \geq 4$ (ii) $x < 4$ (iii) $x < 2$ |
| Q3 | Practice • Easy | (i) $x \leq 2$ (ii) $x > -3$ |
| Q4 | Practice • Medium | $x \geq \frac{9}{2}$ |

IDEA 02 • QUADRATIC INEQUALITIES

- | | | |
|------------|---------------------------------------|-------------------------------------|
| Q5 | Dr Eslam • Inside the roots | $-3 \leq x \leq 4$ |
| Q6 | Dr Eslam • Outside the roots | $x < 1$ or $x > 3$ |
| Q7 | Practice • Easy | (a) 3, 8 (c) $3 < x < 8$ |
| Q8 | Practice • Medium | $x > 1$ |
| Q9 | Practice • Very Hard | $2 \leq t < 5$ |
| Q10 | Pearson IAL WMA11 • June 2023
• Q1 | $x \leq -\frac{1}{4}$ or $x \geq 2$ |

IDEA 03 • INEQUALITIES FOR K

Q11 Dr Eslam • No real roots $4 - 4k < 0 \Rightarrow k > 1$

Q12 Practice • Easy (a) $64 - 16k$ (b) $k < 4$

Q13 Practice • Medium $k < 0$ or $k > 4$

Q14 Practice • Hard $-2 < k < \frac{2}{3}, k \neq 0$

Q15 Pearson IAL WMA11 • January 2020 • Q8 $-2 < k < 5$

IDEA 04 • REGIONS ON A GRAPH

Q16 Practice • Easy $y \geq -1, y \leq x, x < 3$

Q17 Practice • Medium $y < x, y \geq 3 - x, x < 4$

Q18 Pearson IAL WMA11 • June 2023 • Q7 (a) $a = \frac{9}{4}, k = \frac{8}{9}$ (b) $x \geq \frac{9}{4}, y \geq \frac{8}{9}x, y \leq 10 - 2x$

Q19 Pearson IAL WMA11 • June 2024 • Q6 (a) $P(-6, 45), Q(8, 115)$ (b) $y \geq 0, y \leq 5x + 75, y \leq 2x^2 + x - 21$

Q20 Pearson IAL WMA11 • January 2025 • Q9 (a) $\frac{9}{4} - (x-3)^2$ (b) $P(3, \frac{9}{4})$ (c) $x = \frac{9}{2}, k = \frac{\pi}{3}$, period 6 (d) $y \geq \cos \frac{\pi x}{3}, y \leq 6x - \frac{27}{4} - x^2, y \leq \frac{27}{4} - \frac{3}{2}x$

Q21 Pearson IAL WMA11 • January 2026 • Q3 (i) $z < 0$ or $z > \frac{1}{2}$ (ii) shade between the line and the curve for $-2 \leq x \leq -1$ and $x \geq 5$